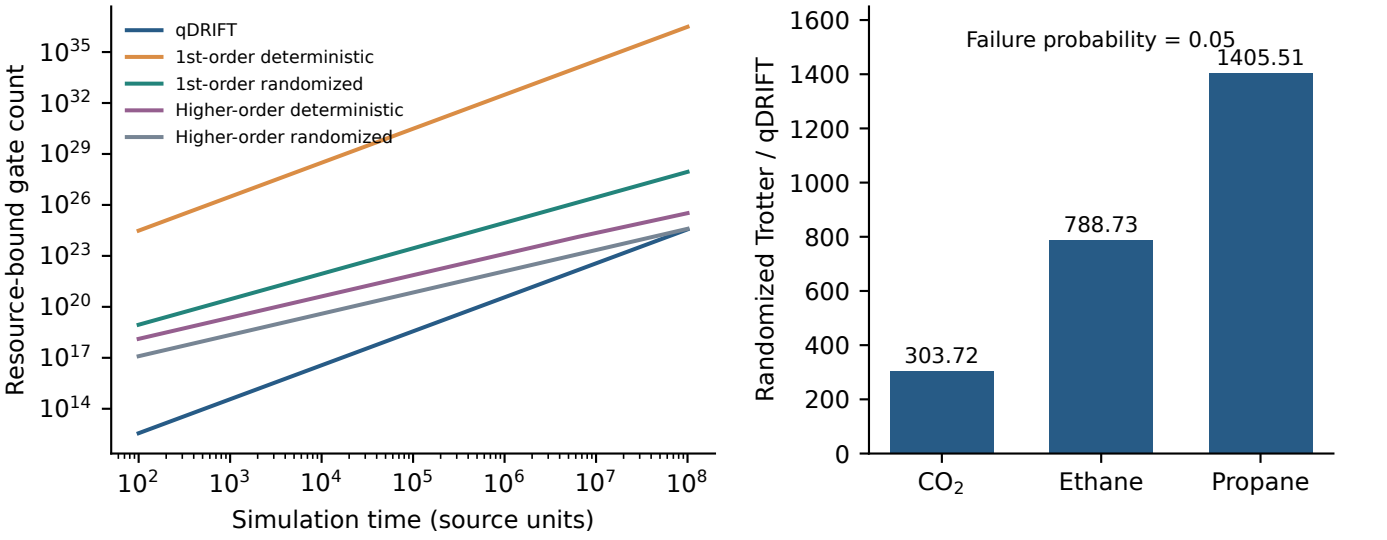
a Propane Hamiltonian resource bounds
qDRIFT
1st-order deterministic
1st-order randomized
Higher-order deterministic
Higher-order randomized
Resource-bound gate count
10^{35}
10^{32}
10^{29}
10^{26}
10^{23}
10^{20}
10^{17}
10^{14}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
10^{8}
Simulation time (source units)
b Phase-estimation resource ratio
Failure probability = 0.05
Randomized Trotter / qDRIFT
1600
1400
1200
1000
800
600
400
200
0
1405.51
788.73
303.72
CO_2
Ethane
Propane